FINAL JAVASCRIPT EXERCISES (30 REAL-LIFE PROBLEMS)

1. Age Verification System

You are building a system for a voting center.

Create a variable called `age`. Your program must decide if the person is allowed to vote.

If the age is less than 0, show “Invalid age." If it is less than 18, show that the person cannot vote. If it is between 18 and 59, allow voting. If it is 60 or above, allow voting and show a message for senior citizens.

Make sure your conditions are correct and do not overlap.

2. Temperature Clothing Advisor

You are creating a simple weather assistant.

Create a variable `temperature`. Based on the value, suggest what the user should wear.

If the temperature is below 15, suggest warm clothes. If it is between 15 and 25, suggest normal clothes. If it is above 25, suggest light clothes.

If the value is unrealistic (like extremely low or negative beyond normal), show an error.

3. Student Grade Analyzer

You are building a school grading system.

Create a variable named `marks`. The value must be between 0 and 100.

If the value is outside this range, show “Invalid marks." Otherwise, assign a grade: A, B, C, D, or F. If the marks are below 50, the student fails. Otherwise, they pass.

Also display a short message like "Excellent," "Good," or “Needs improvement."

4. Bank Withdrawal System

You are building part of an ATM system.

Create variables `balance` and `withdrawAmount`.

If the withdrawal amount is less than or equal to 0, show an error. If it is greater than the balance, show “Insufficient funds”. Otherwise, subtract the amount using `balance -= withdrawAmount` and display the new balance.

5. Login System with Limited Attempts

You are building a secure login system.

Store a correct username and password in variables.

Allow the user to try logging in, but only up to 3 times. Use a loop to repeat the attempts.

If the user enters correct details, show success and stop the loop using `break`. If they fail 3 times, show that the account is blocked.

6. Day Planner (Using Switch)

You are building a weekly planner.

Create a variable `dayNumber` (1–7). Use a switch statement to display the correct day and suggest an activity.

If the number is not between 1 and 7, show an error.

7. Simple Calculator (Using Switch)

You are building a calculator.

Create variables `num1`, `num2`, and `operator`.

Use switch to perform the correct operation. Handle addition, subtraction, multiplication, and division.

If the operator is invalid, show an error. If the user tries to divide by zero, prevent it.

8. Traffic Light System

You are programming a road system.

Create a variable `color` and another variable `isEmergency`.

If the color is red, yellow, or green, show the correct action. If it is an emergency vehicle, allow it to move even if the light is red.

9. User Role Access System

You are controlling access to a system.

Create a variable `role`.

If the role is admin, give full access. If it is user, give limited access. If it is guest, allow only viewing. If it is anything else, show an error.

10. Restaurant Menu Selector

You are building a digital menu.

Create a variable `choice`.

Each number should represent a food item and price. Display the selected item. If the number does not match any item, show an error.

11. Countdown Timer

You are creating a countdown for an event.

Use a loop to count from 10 down to 1. Print each number. After the loop finishes, print “Start”.

12. Student List with Stop Condition

You are printing student names.

Create an array of names. Loop through the array and print each name.

If the name “Admin” appears, stop the loop immediately using `break`.

13. Multiplication Table Generator

You are helping students learn multiplication.

Create a variable `number`. Use a loop to print its multiplication table from 1 to 10.

14. Even Number Finder

You are filtering numbers.

Use a loop from 1 to 50. Skip all odd numbers using `continue`. Only print even numbers.

15. Shopping Cart Total

You are building an online shopping system.

Create an array of prices. Use a loop to calculate the total using `total += price`.

If any price is negative, skip it using `continue`.

16. ATM Simulation System

You are building a full ATM system.

Create a menu that keeps running using a loop. The user can choose to check balance, deposit, withdraw, or exit.

Use switch for the menu. Use `break` to exit the system. Make sure withdrawals do not exceed balance and deposits are not negative.

17. Password Attempt Limiter

You are securing a system.

Allow a user to enter a password only 3 times. If they enter the correct password, stop immediately. If they fail all attempts, block acces.

18. Bus Ticket Price Calculator

You are building a transport system.

Create variables `age` and `distance`.

If age is invalid (negative), show error. Children pay less, adults pay normal price, seniors get discounts. If distance is long, apply extra discount.

19. Quiz Game

You are building a quiz system.

Store 5 questions and answers. Use a loop to go through them.

If the answer is correct, increase score using `score += 1`. If the answer is empty, skip it using `continue`.

20. Online Store Discount System

You are building an e-commerce system.

Create variables `amount` and `customerType`.

If amount is high and customer is VIP, give a large discount. If only one condition is true, give a smaller discount. Otherwise, no discount.

21. Number Comparison Tool

Create two numbers.

Compare them and display which one is greater. If they are equal, display that they are equal.

22. Number Range Checker

Create a number.

Check if it is between 10 and 50. If yes, show “Within range”. Otherwise, show “Out of range”.

23. Odd Number Skipper

Loop from 1 to 30.

Skip all odd numbers using `continue`. Print only even numbers.

24. Stop Loop at Target

Loop through numbers.When the number reaches 20, stop the loop immediately using `break`.

25. Sum of Numbers

Calculate the total from 1 to 100.

Use a loop and add values using `total += i`.

26. Password Strength Checker

Create a password variable.

Check if it has at least 6 characters, contains a number, and is not empty.

Use logical operators like `&&`.

27. Email Validator

Create a variable `email`.

Check if it contains both “@” and “.”. If yes, it is valid. Otherwise, invalid.

28. Product Stock Checker

Create a variable `stock`.

If it is 0, show out of stock. If less than 5, show low stock warning. Otherwise, show available.

29. Average Marks Calculator

Create an array of marks.

Loop through the array and calculate the average. If any value is invalid, skip it using `continue`.

30. Game Score Tracker

You are building a simple game.

Use a loop to simulate rounds. Add points using `score += points`.

If the score reaches a target value, stop the loop using `break`.

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The Chief HR & Administration Officer
RwandAir Limited
Kigali, Rwanda

Dear Sir/Madam,

RE: APPLICATION FOR QUALITY MANAGER FLIGHT OPERATIONS POSITION

I am writing to apply for the Quality Manager Flight Operations position advertised on the RwandAir website. As an experienced Aircraft Maintenance Engineer, I bring strong expertise in aviation safety, regulatory compliance, and quality assurance that directly aligns with this role's requirements.

My background has equipped me with comprehensive knowledge of aviation regulations (ICAO standards and civil aviation requirements), quality management systems, and operational safety protocols. I have developed strong skills in implementing corrective actions, and collaborating across flight operations, maintenance, and safety departments to ensure the highest standards of excellence.

I am particularly drawn to RwandAir's reputation as East Africa's leading carrier and your commitment to operational excellence. I am confident that my technical expertise and dedication to safety will enable me to contribute effectively to your continued success and growth.

I have enclosed my curriculum vitae for your consideration and would welcome the opportunity to discuss how my qualifications align with RwandAir's objectives.

Thank you for considering my application. I look forward to hearing from you.

Yours faithfully,

Byagatonda Joseph